

"It was Empowering": Care Pathways a Novel Culturally Informed Inspiration Card Game for Community-Engaged Co-Design with Black American Women and Clinicians

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Abstract

Anxiety and depression are the most common mental health conditions among Black American women. Despite the prevalence of these mental health conditions, gendered racial/ethnic treatment disparities persist. Culturally responsive digital mental health applications represent one promising path for expanding access to evidence-based mental health care. However, Black American women are underrepresented in the design and development of digital mental health applications. In this work, we report preliminary findings from four co-design workshops with seven Black women with a history of anxiety and/or depression and seven mental health professionals. We share with the community our novel inspiration card game—*Care Pathways*—used to engage participants in the co-design of a digital mental health application that leverages culturally responsive Mindfulness-Based Cognitive Therapy. Our findings examine how *Care Pathways* sparked meaningful and rich design insights while supporting two sets of experts, Women with lived experience and mental health professionals, in co-designing together.

CCS Concepts

• **Human-centered computing** → **Participatory design**; *User centered design*.

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1 Introduction

In the United States, anxiety and mood disorders (eg, major depressive disorder) are the most common mental health conditions among Black women [15]. Black American women are underdiagnosed, diagnosed later, and endure more severe and ongoing symptoms of distress than women from other racial and ethnic demographics [21, 39]. This is partly due to a confluence of socio-ecological factors that disproportionately affect Black women, including racial and gender discrimination and a perceived obligation to adhere to societally imposed cultural archetypes (i.e., Black Superwoman Schema [44], Strong Black Woman [42]), higher exposure to violence, socioeconomic disparities, and limited access to healthcare [15, 32]. Recent research indicates that younger Black American women aged 25-34 experience a higher severity and chronicity of anxiety and depression compared to their older counterparts, putting them at particular risk [24, 25].

Despite the prevalence of depression and anxiety among Black American women, they underutilize professional mental health treatment compared to women from other racial and ethnic groups, with some estimates suggesting that only 7% of Black American women experiencing depression are engaged in treatment [5, 42]. This undertreatment stems from socio-cultural and systemic barriers present in American society, including justified cultural mistrust of providers, a lack of identity-congruent providers, stigma, and structural inequities [5, 27].

Digital mental health tools have shown promise in expanding evidence-based care, increasing access to specialized providers through telehealth [2, 22], and addressing structural barriers that hinder treatment participation [17, 20]. Despite the numerous benefits, Black American women remain at the margins of digital mental health technology co-design research, resulting in their specific needs, challenges, preferences, and experiences being largely overlooked.

Our work explores this pathway and is part of a multi-year project to engage Black women ages 25-34 with elevated anxiety and/or depression and mental health professionals in the design and evaluation of a culturally responsive digital mental health application.

We report preliminary findings from four early phase co-design workshops with seven Black women with a history of anxiety and/or depression and seven mental health professionals to present our novel co-design method—*Care Pathways*. Design workshops were conducted to explore opportunities for a smartphone-based intervention to support the self-management of anxiety and depression among Black women through culturally responsive Mindfulness-Based Cognitive Therapy (MBCT) [36]. We provide findings of our evaluation of *Care Pathways* to demonstrate its effectiveness in facilitating two groups of experts—subject matter experts and experts by experience—to co-design together, developing rich and culturally informed digital mental health design insights. The goal of this work is to introduce the *Care Pathways* inspiration card game and materials to the Human-Computer Interaction (HCI) community to spark discussions and advance the practice of community-engaged co-design with multiple stakeholders from traditionally excluded communities. Our work examines these research questions:

RQ1: How might the challenges, microaggressions, and symptoms of distress commonly experienced by Black women shape their desires for therapeutic content and features in a digital mental health app?

RQ2: How should Mindfulness-Based Cognitive Therapy (MBCT) techniques be culturally adapted to address the mental health needs of Black women in a digital mental health app?

RQ3: How might the culturally informed inspiration card game *Care Pathways* engage and support subject matter experts and experts by experience in co-designing digital mental health tools together?

2 Related Work

2.1 Participatory Design and Inspiration Cards

HCI and digital health researchers have a rich history of integrating participatory design and community-engaged research methods (e.g., Community-Based Participatory Research [38], Asset-Based

Design [1, 3], Community-Based Participatory Design [8], etc.) to design, develop, and evaluate digital health interventions with their intended users. The strengths of this work include employing community-engaged and Participatory Design (PD) methods to reveal structural inequities that hinder communities from realizing their health goals, building rapport, and providing participants ownership over project direction and outcomes. PD has successfully created digital health technologies that honor and are responsive to user values, goals, norms, and beliefs [30, 33, 46]. Envisioned as design workshops, participants gather with researchers to co-create using design activities (e.g., storyboards [35, 37], Photovoice [18, 29], etc.) that stimulate conversation and explore concepts for design.

Inspiration cards are one method used to conduct exploratory co-design work with participants [6] where participants are given a deck of cards with multiple card types (i.e., domain and technology cards) to match and combine [6]. The resulting combinations offer early theoretically driven design concepts of digital systems assembled by participants. Inspiration cards have catalyzed meaningful involvement by participants, producing rich design ideas [12, 40]. The creation of a culturally informed digital mental health application is a complex undertaking requiring designers to have knowledge of theories that influence mental wellness (i.e., cultural schemas, stigma), techniques from psychology (i.e., MBCT), and experience with different technologies (i.e., AI, online health communities). Yet, it remains crucial that technology systems designed to alleviate health disparities include community members at every stage of research and design. Responding to this need, we created a set of theoretically driven inspiration cards to scaffold participant brainstorming.

2.2 Digital Mental Health with Black Women

Digital health research on the design and evaluation of evidence-based digital mental health technologies with Black women is a burgeoning area of research. To date, digital mental health survey research has focused on building a literature surrounding Black women's attitudes and preferences for digital mental health interventions focusing on depression, generalized anxiety, and postpartum depression [24, 28]. Findings suggest a strong inclination by Black women to adopt mobile technology to access mental health providers [24] and social networks [28], early studies have signaled acceptance of digital technologies that promote self-care [23, 26] and prioritize social connection [10].

Although Black women have a positive perception of using mobile technology to manage mental health [9, 13, 24, 28], usage data indicate that existing digital health tools are failing to provide adequate care [9]. We extend related work by exploring how to *collaboratively* develop an MBCT application with Black women and mental health professionals to ensure that content is professionally informed and culturally relevant.

3 Method

3.1 Designing *Care Pathways*: A Culturally Informed Inspiration Card Game

Care Pathways is an inspiration card game (Fig 1). The game's goal is to create a pathway to care by combining different card types from

the *Care Pathways* card decks, setting them on the gameboard, and reflecting on design decisions using an accompanying worksheet. We designed a set of cards with five subtypes: (1) situations, (2) techniques, (3) supporters, (4) technologies, and (5) idealized futures cards. The focus of the activity was to explore the *situations* Black women experience and the *idealized futures* they may want to see manifested in combination with possible MBCT *techniques*, interactions with preferred *supporters*, and desired *technologies* to promote Black women's mental health. Each inspiration card represented a concept. The front of the card had a few words representing the concept. The back had a few sentences defining the concept. Each card introduced concrete yet flexible concepts for participants to leverage when brainstorming features.

3.1.1 Five Card Types. *Situation* cards included 46 cards defined as challenges, microaggressions, or symptoms of distress commonly experienced by Black women. These cards were theoretically driven, leveraging concepts and dimensions from cultural schemas and frameworks that identify unique racialized gender norms (e.g., Black Superwoman Schema [45]) and manifestations of symptoms of anxiety and depression described by Black women [16]. The *Technique* cards included 15 cards representing evidence-based therapeutic practices directly from MBCT. MBCT is a therapeutic approach that combines elements of mindfulness training (e.g., sitting meditation, Body scan, breathing exercises, etc.) with techniques from Cognitive Behavioral Therapy (e.g., reframing). Using these techniques, individuals experiencing depression and/or anxiety can develop cognitive skills, such as self-awareness, to relate differently to their physical sensations, thoughts, and feelings. These skills help individuals disrupt harmful cognitive processes that underlie depression such as rumination [36]. *Supporter* cards included 16 cards representing the people the women would like to do the technique with, learn from, or support. *Technology* cards included 15 cards featuring technologies used to receive, practice, and experience the technique and/or interact with preferred supporters. *Idealized future* cards included 25 cards representing the outcome, experience, or feeling Black women hoped to create or augment for themselves or their clients using the *Care Pathway* they brainstormed. Card and gameboard wordings were tailored slightly for both experts.

After developing the cards and gameboard for *Care Pathways*, we met with subject matter experts to ensure the game's content was culturally responsive. We also conducted a mock workshop with research affiliates unfamiliar with the design activity and integrated their feedback.

3.1.2 Game Mechanics. As designed, *Care Pathways* is a single-player card game. Each player is given five card decks and a game board (Fig 1). The gameboard has five sections; each section includes a step instructing the player to choose cards from the corresponding card deck and place those cards on the board. In step 1, the player is instructed to choose 1-3 situation cards from their situation card deck and place those cards on the gameboard. In step 2, the player picks 1-2 technique cards that they believe could help with the situations from step 1. In step 3, the player picks cards from the supporter deck to represent who they would like to interact with when engaging in the techniques from step 2. In step 4, the player picks technology cards to represent how they might like to

experience the techniques they chose in step 2 using technology. In step 5, the player picks future cards to represent the future outcome they hope to achieve or experience. By completing all five steps, the player has assembled their own unique pathway to care from situation to outcome using cards from the five decks. Each player's *Care Pathway* was used as a probe to prompt discussion between workshop participants. For exact wordings, please refer to the gameboards in our supplementary materials.

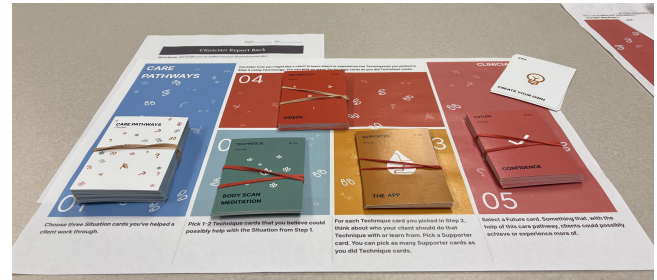


Figure 1: *Care Pathways* inspiration card game, including the gameboard, five card types, a *create your own* card, and worksheet

3.2 Workshop Groups

When designing digital mental health tools, it is critical to get design guidance from both experts by experience (i.e., women with depression and/or anxiety) and subject matter experts (i.e., licensed mental health professionals). We believe there is value in supporting co-design between these two experts, allowing them to develop design insights in the room together, rather than conducting parallel workshops with single expert groupings that rely on researchers to integrate experts' feedback in the analysis phase only. We wanted the workshops we conducted to include opportunities for both types of experts to react and engage in meaningful dialogue about their designs with each other directly. One aim of these data collections was to investigate the benefits and potential drawbacks of co-design workshops with two types of experts compared to those with a single expert in attendance. For example, we wanted to explore if potential power dynamics between experts by experience and subject matter experts changed how participants interacted with one another during the workshops. To make these comparisons, we ran four workshops with the following expert structure: two mixed workshops (Black women with lived experience and mental health professionals), one workshop with Black women with lived experience only, and one workshop with mental health professionals only.

3.3 Workshop Procedures

The study was approved by our university's Institutional Review Board. Black women with a history of anxiety or depression were recruited via posts on social media and fliers. Black women (ages 25-34 years) with lived experience, defined as either having (a) diagnoses of anxiety or depression; (b) elevated anxiety (score ≥ 10) on Generalized Anxiety Disorder 7-item scale (GAD-7)) [34]; elevated depression (≥ 10 on Patient Health Questionnaire 8-item scale

(PHQ-8) [14] or (c) history of experiencing symptoms of anxiety or depression were eligible to participate. Mental health professionals were recruited by using therapist directories and emailing them an invitation to participate and disseminating fliers. Participants were invited to one of four workshops (3–4 participants per session). Electronic Consent was obtained for all participants before study participation.

At the beginning of the session, the last author introduced the project and gave an overview of MBCT. During the session, the lead author provided an overview of how to complete a care pathway and participants were asked to complete a demographic survey. Participants were then asked to build their care pathway(s) and complete their accompanying worksheet. Next, the lead author facilitated the discussion with participants sharing their care pathways by discussing their card combinations. The participants then completed the Rose-Bud-Thorn activity [11, 43], an open-ended survey featuring three questions focused on evaluating the *Care Pathways* activity, asking participants to identify (1) the positive aspects (Rose), (2) the promising aspects (Bud), and (3) the problematic aspects (Thorn) of the activity. At the end of the workshop, each participant received a \$100 gift card.

3.4 Analytic Approach

Design workshop discussions were audio recorded and transcribed. We conducted an inductive thematic analysis using the Sort, Sift, Think, and Shift analytic approach [19]. First, project members independently read transcripts and unstructured survey responses, reviewed participant care pathways, wrote analytic memos, identified powerful quotations, and met weekly to discuss. Next, members created a list of core topics to code and monitor actively. Project members reviewed inventories of their most powerful quotes across participants and their coded data—bridging topics of meaning and relevance to identify and develop core themes.

4 Findings

We report three themes from the design workshops based on participants' discussions of their cards and a method evaluation of *Care Pathways*. Quotations are annotated with CM or CL to refer to their expert status as Black women with lived experience, community members (CMs), or mental health clinicians (CLs). Seven Black women with a history of anxiety and/or depression and seven licensed mental health professionals participated in the workshops. All participants self-identified as Black women.

4.1 Theme 1: Safe Digital Mental Health Spaces that Validate and Support Black Women (RQ1)

Participants expressed a deep desire to be genuinely recognized, stemming from experiences that result from gendered and racialized disenfranchisement. They believed one way to achieve this was by accessing safe digital spaces that fostered praise and acknowledgment within a community of Black women. P5(CL) voiced, *"I just need something in there to keep the person going... Look, the app cares....Even with other Black women, sometimes we're like, I see you, Sis, I see you."* Black women from the group explained how their own needs come last due to the societal demands and

cultural expectations of caring for others disproportionately placed on Black women in their personal and professional lives. P11(CM) shared, *"So it's that Black women, they don't have a sense of belonging because we work and live in White spaces...there aren't safe spaces."*

In their professional lives, they recounted experiences of feeling unsafe, unsettled, and uncomfortable, experiences rooted in institutionalized practices of racism that affect their daily lives, while discussing the tensions between their idealized future and situation cards. P4(CM) shared the paradox she sees between the idealized future she would like and the external factors that prevent it: *"Letting down one's guard [future card] in these environments can feel really unsafe. I want to have ease [future card], but I also want to speak up for myself."* P2(CM) recounted experiences that caused her distress in the workplace. She described how the Strong Black Woman archetype had been imposed on her by white colleagues. *"I keep hearing, 'You got this, you can handle it.' So, it's like, why do you think I have to be strong? It always happens in the workplace. All higher up, they are White. I'm telling you I need help, I need help...they don't realize it until another card 'tone'. Your tone changes."*

The microaggressions and frustrations voiced by participants, along with their future and situation cards, demonstrated that for digital mental health platforms to be *truly* culturally responsive, they must respond to these unsafe spaces by fostering community and helping Black women feel understood, accepted, and supported by other Black women to ultimately *be seen*. Participants selected a combination of supporter and technology cards to represent features for their idealized digital communities, starting with the need for (1) peer support, (2) access to mental health professionals, (3) celebratory spaces, and (4) discretion.

4.2 Theme 2: Stop the Spiral and Focus on Self-Awareness with Low-Burden Grounding Techniques. (RQ1 & RQ2)

Participants added context to the *aches and pains* situation card, discussing various somatic concerns they felt throughout their bodies in connection to other situations. This was often done in combination with cards focused on chronic stress, ruminations, or worries over past or future events. By selecting situation cards, participants identified a lack of self-care and time for themselves, compounded by additional situation cards: life transitions, family stressors, and the pressure to succeed. P5(CL) voiced the competing demands Black women juggle, *"You got to pause, especially women, it is just go, go, go. Holding down a household, holding down work, being a wife, running businesses."* The experiences, represented by the situation cards, were followed by mindfulness technique cards for in-app activities that participants believed could *stop the spiral* by encouraging them or their clients to take *at least a minute for themselves and breathe*. The women described these MBCT activities as grounding techniques—a means to center oneself.

In addition to grounding MBCT techniques, participants requested features that present objective data and app-based interpretations of their historical data for self-reflection. They described the goal of these features as highlighting progress. P9(CM) shared her reasoning, *"Historical content [technology card] to show progress*

because sometimes it feels like especially when you're taking a step back, and you have mental health issues, it can send you down a dark rabbit hole of all kinds of intrusive thoughts. So, seeing...these are things that worked, and I made progress here. I'm not all the way back where I started."

Participants were open to using the app to celebrate their incremental progress and identify achievements that might otherwise go unnoticed during times of distress. P5(CL) described how she envisioned historical content benefiting users by combining the technology card with a technique card, *"I did the historical content [technology card]. I would love it if you used the journaling [technique card] and you had a bad day, but then you talked about how you overcame that and what techniques you used. Then you have a database, so you visibly see your growth"* Feedback and card selections showed the importance of digital mental health tools designed to cultivate user insights centered on personal growth, a core component of MBCT, by explicitly acknowledging coping mechanisms.

4.3 Theme 3: Culturally Responsive Digital Mental Health Tools that Shift Users' Perspectives to Take Back Control and Move Forward. (RQ1 & RQ2)

Participants envisioned application features that would immerse them, shut out current stressors, reconnect with their bodies, and reorient to the present to help identify their feelings and assert more control over their responses. P8(CL) voiced, *"And then journaling [technique card]. I can do a brain dump, just get everything out, and then I can breathe in like, alright, what's the next path for me? And then this way, right? No, this way."* Aligning with MBCT's focus on emotion regulation, MBCT techniques were frequently discussed by participants in relation to the need to self-regulate one's emotions to respond purposefully to environmental and interpersonal stressors. The importance the participants placed on regulating emotions was frequently discussed in the context of gendered racial microaggressions, such as the tone situation card, where Black women are often stereotyped as angry when self-advocating in spaces dominated by White people, revealing a state of constant hypervigilance.

Cognitive reframing involves restructuring negative thought patterns and is a core component from both Cognitive Behavioral Therapy and MBCT. Grounded in Western colonial perspectives, reframing has historically been modeled on the experiences of White upper-class men [4, 31]. When applied to the experiences of marginalized individuals, particularly Black women, reframing can be reductionist and counterproductive. All Black women clinicians believed cognitive reframing should be part of the application while considering certain caveats. P12(CL) shared what reframing looks like in her practice, *"a lot of the Black women I work with who are like, well, I got a promotion, but it's not enough. And I'm like...you're using this to build up to another promotion. It's more of a how does this contribute to yourself right now or to yourself in the future? That has been helpful around microaggressions...I can't be like, we can reframe the racism you've experienced."* P12's quotation highlights this notion that

reframing exercises done without a culturally responsive and historical lens can invalidate Black women's realities by implying that the impact of their experiences is their responsibility and merely a matter of reinterpretation. P12's quote demonstrates how cognitive reframing can adopt a mindful approach by reframing a negative thought to reduce its emotional distress.

4.4 Method Evaluation: Care Pathways Supported Intuitive Co-Design between Experts. (RQ3)

All participants reported that *Care Pathways* was engaging and intuitive. Ultimately, the inspiration cards and worksheet served as design probes—a flexible set of materials aligned with a specific research goal whose contents were designed to elicit inspirational responses from participants using creative means [41]. Participants appreciated how the cards and gameboard helped scaffold their feedback. One participant noted, *"I love the idea of having cards to scan quickly to dive into something like this. It was empowering."* Additionally, participants appreciated combining card types to co-create unique design opportunities in the context of their lives. One participant shared in the exit survey, *"The activity itself is so well designed...It was a helpful thinking tool for me to strategically align concerns with potential tech options."*

Care Pathway's provided equal footing and allowed for both sets of participant experts to design together. Our approach and the welcoming atmosphere of our session helped mitigate potentially harmful power dynamics that could have inhibited co-creativity. In our observations of the groups with both sets of experts, we witnessed the same level of encouragement, engagement, curiosity, and camaraderie as we did in each single expert session, displayed as laughing, smiling, and words of praise. Furthermore, the inspiration card types—techniques and technologies—assisted both experts in familiarizing themselves with new techniques or technologies they might otherwise not have known or forgotten. Although our expert participants' perspectives overlapped, they were substantially different. The cards seamlessly provided a foundation of knowledge for participants to draw on and add to.

4.5 Method Evaluation: Care Pathways was Culturally Informed and Adaptive. (RQ3)

Participants' perspectives revealed that the *Care Pathways* game was culturally responsive through its use of (1) culturally informed materials and (2) supported participant-controlled disclosures of personal narratives. When reflecting on the situation cards, participants stated that they appreciated discussions on the external factors and historical influences that contribute to Black women's poor mental wellness. One participant voiced, *"The cards were so thoughtful and representative of Black women's experiences."* Another stated, *"There were things that were very specific to Black women...like, social justice expert [situation card], too visible [situation card], not visible enough [situation card]. All of those cards were very appropriate."* All participants stated that the game was representative of Black women's experiences.

When designing inspiration cards, designers aim to keep the decks small. Yet, when it came to representing culturally informed symptoms of distress, microaggressions, and cultural expectations

that influence Black women’s experiences of depression and anxiety, it was clear our situation card deck would be extensive (46 cards). Even still, these situations were not exhaustive. Anticipating the need for participants to expand these decks, we included a *Create Your Own* card. Using these, Participants identified four additional situations: (1) finances, (2) divorce, (3) single parenting, and (4) trauma. One clinician expanded the technique deck to include a new technique, and two others created a spiritually augmented set of MBCT techniques, encouraging religious participation and spiritual practice.

Related work has urged researchers to consider ways to make research spaces safer for participants by minimizing the need to share personal details that may be triggering [7], prioritizing participant well-being over detailed narratives. When using *Care Pathways* to connect contexts (i.e., situations) to techniques, both sets of experts had agency over the number of personal details they shared by relying on the cards to introduce topics instead of grounding them in personal narratives. Some participants shared their reasons for their cards, others kept the details private, and others chose to elaborate on their selections in writing, supporting controlled self-disclosure.

4.6 Method Evaluation: *Care Pathways* Fostered Opportunities for Fellowship and Connection. (RQ3)

A design session focused on sensitive topics such as mental illness and gendered racial inequities could be emotionally draining. An observation shared by our participants indicated in an exit survey: ***“I like that there was still a light tone to the session despite discussing dark topics, which made me feel more comfortable being vulnerable.”*** Although the participants had never met before, a sense of connection and shared understanding permeated the discussions. We witnessed participants smiling, laughing, nodding, and singing along to music while co-designing together. At the end of the workshops, we saw participants engaging in smaller chats, exchanging contact information, and lingering well beyond the two hours after the workshop to discuss the project in community with each other and us.

All the exit surveys included Roses of the session centered on the support and sense of caring participants found in this welcoming group setting with other Black women. Participants shared these Roses: ***“Building a relationship with the other Black women.”***, ***“It felt like fellowship.”*** and ***“Sitting together with other Black women and hearing how they saw their boards was thought-provoking.”*** Participants found meaning and comfort in the common threads they shared, with many women noting similar card selections from the deck of 46 situation cards. One exit survey stated, ***“The discussion was amazing, everyone’s natural reactions to similar topics and open-mindedness.”*** Another survey noted the benefit of sharing in a group setting, ***“I enjoyed talking through my care path and listening to what others had to say about theirs. It’s one thing to go through the cards. Sharing with others added more context to my own care path.”*** The participants recognized the benefit of completing the activity together, adapting their pathways based on others’ insights.

5 Discussion

When it comes to digital health intervention co-design, *Care Pathways*’ cards and gameboard templates are flexible and can be adjusted to suit a multitude of project needs (e.g., including fewer card decks to represent fewer domains, adapting to new populations and health topics, adding cards that focus on intersectionality, tailoring language on the boards based on project goals, adjusting reflection worksheets, etc.). Researchers could utilize the *Care Pathways* game in consensus workshops or use *Care Pathways* as a cultural probe, later grounding individual interviews. We believe that changes to board size and cards and adaptations to the worksheets to add context would be what’s needed to cleverly adapt the method.

Care Pathways played a crucial role in fostering a welcoming, engaging, and productive atmosphere for a workshop where two groups of experts from historically marginalized communities—Black women with lived experiences of anxiety and/or depression and mental health professionals—could come together in community and co-create. As researchers, we strive to build engaging and creative data collections that give participants ownership and agency over the research process to ultimately effect positive social change within their communities based on their design insights. Beyond the traditional long-term benefits of research participation, our participants highlighted the near-term benefits they experienced, finding value in the safe space of the workshop surrounded by other Black women.

6 Conclusion

As a culturally informed inspiration card workshop and design probe, *Care Pathways* engaged participants creatively and garnered design insights for culturally responsive MBCT techniques embedded within a digital mental health application. We plan to update the application’s design and invite participants to a community-engaged dissemination session. We found that the design and the session format served as a mechanism for catalyzing meaningful participant engagement, building a sense of community, and supporting interactions between Black women with lived experience and mental health professionals. Our work serves as an example of how to construct a theoretically driven and culturally responsive inspiration card workshop in service of co-designing digital mental health applications with historically marginalized populations.

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